Intangible pollutants

Pollution in the form of energy, be it light, noise, or heat, can be just as intrusive as physical waste or chemical emissions. Light pollution from buildings, streetlights, vehicles, and advertising billboards was first described as a problem in New York in the 1920s. It can cause problems for nocturnal wildlife, for example, because predator–prey relations are interrupted. Excessive noise can be highly disturbing in cities, on flight paths, and near factories and roads. But it also affects wildlife in subtler ways. There is evidence that some birds now sing at night because their song can be heard more clearly than during the day.

Waste heat, too, can be damaging. When water from rivers or the sea is used as a coolant in factories or power stations, the hot water that is returned to the source is a form of thermal pollution. It can alter the composition of the plant and animal species living in the area.

Nuclear energy is sometimes viewed as “cleaner” than fossil-fuel energy, because it does not produce greenhouse gases, but it does result in waste that remains radioactive for thousands or millions of years. The industry also bears the inherent risk of accidental damage. An explosion at the Chernobyl nuclear power plant in Ukraine in 1986 killed dozens of people and spread radiation across Western Europe. The slowly dwindling effects of contamination on the ecosystem and human health are predicted to last a century.

“In 2015, pollution caused three times as many deaths as AIDS, tuberculosis, and malaria combined.”

Philip Landrigan